

读法 | 每格左半是源码，右半是这段源码渲染出的图，两者由同一份代码生成，不会脱节。标题右侧灰字标出该格的关键选项与所属库，代码中出现的库需先 `\usetikzlibrary{...}`。

全局约定 | 本文档设了 `\tikzset{every picture/.append style={font=\scriptsize}}`，故图内文字以默认小一号；除此之外均为 TikZ 原生行为，代码可直接复制。

五个基本命令 | `\path` 只构造路径不作任何绘制，是一切的基础；`\draw=\path[draw]` 描边，`\fill=\path[fill]` 填充，`\filldraw` 两者兼有，`\shade` 渐变；`\node/\coordinate/\pic` 同样都是 `\path` 的简写。

覆盖 | 全表 55 格，依次是基础语法、节点系统、结构化编程、曲线与装饰、经典成图，末尾附库清单、十个常见陷阱与编译导出命令。

55 格索引 点击条目跳到对应格，pN 为页码；书签栏与页脚的「索引」也可导航

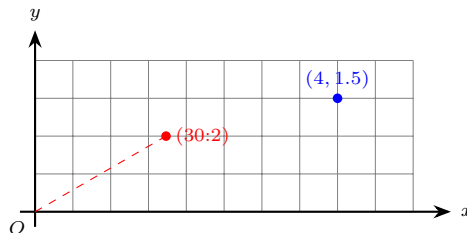
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一 · 基础语法

坐标 / 路径 / 线型 / 颜色 / 箭头 / 填充 / 变换

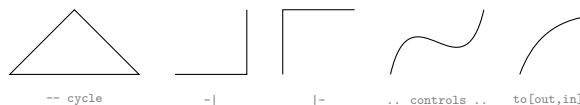
1 坐标系 笛卡尔 · 极坐标 · x/y 单位向量

```
\begin{tikzpicture}[x=1cm,y=1cm] % unit vectors
\draw[help lines,step=.5] (0,0) grid (5,2);
\draw[thick,-Stealth] (-.2,0) -- (5.5,0) node[right]{$x$};
\draw[thick,-Stealth] (0,-.2) -- (0,2.4) node[above]{$y$};
\fill[blue] (4,1.5) circle(1.8pt) node[above]{$(4,1.5)$};
\draw[red,dashed] (0,0) -- (30:2); % polar = (angle:r)
\fill[red] (30:2) circle(1.8pt) node[right]{$(30{:}2)$};
\node[below left] at (0,0) {$0$};
\end{tikzpicture}
```



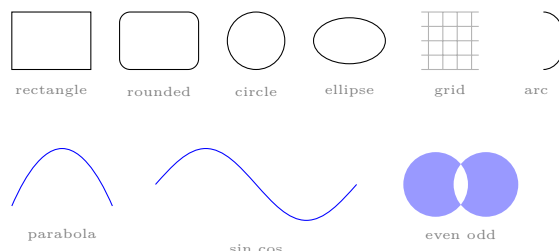
2 路径运算符 -- -| |- controls to cycle

```
\begin{tikzpicture}[scale=.95]
\draw (0,0) -- (.9,.9) -- (1.8,0) -- cycle;
\draw (2.3,0) -| (3.3,.9);
\draw (3.8,0) |- (4.8,.9);
\draw (5.3,0) .. controls (5.6,1.3) and (6.3,-.4) .. (6.6,.9);
\draw (7.1,0) to[out=70,in=190] (8,.8);
\foreach \x/\t in {9/{-- cycle}, 2.8/{-|}, 4.3/{|-},
5.95/{.. controls ..}, 7.5/{to[out,in]}}
\node[font=\tiny,text=gray] at (\x,-.35) {\texttt{\t}};
\end{tikzpicture}
```



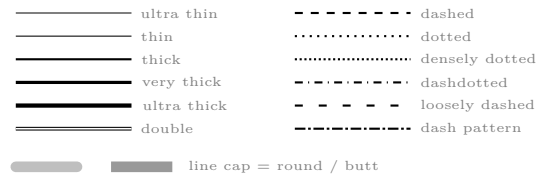
3 基本形状 rectangle · circle · ellipse · arc · grid

```
\begin{tikzpicture}[scale=.95]
\draw (0,0) rectangle (1.1,.8);
\draw[rounded corners=4pt] (1.5,0) rectangle (2.6,.8);
\draw (3.4,.4) circle(.4);
\draw (4.7,.4) ellipse(.5 and .32);
\draw[step=.2,gray!60] (5.7,0) grid (6.5,.8);
\draw (7.4,.8) arc (90:-90:.3 and .4); % arc(from:to:rx and ry)
\draw[blue] (0,-1.9) parabola bend (.7,-1.1) (1.4,-1.9);
\draw[blue] (2,-1.6) sin (2.7,-1.1) cos (3.4,-1.6)
sin (4.1,-2.1) cos (4.8,-1.6);
\fill[blue!40,even odd rule] (5.9,-1.6) circle(.45)
(6.6,-1.6) circle(.45);
\foreach \x/\y/\t in {.55/-./3/rectangle, 2.05/-./3/rounded,
3.4/-./3/circle, 4.7/-./3/ellipse, 6.1/-./3/grid, 7.3/-./3/arc,
.7/-2.3/parabola, 3.4/-2.5/{sin cos}, 6.25/-2.3/{even odd}}
\node[font=\tiny,text=gray] at (\x,\y) {\t};
\end{tikzpicture}
```



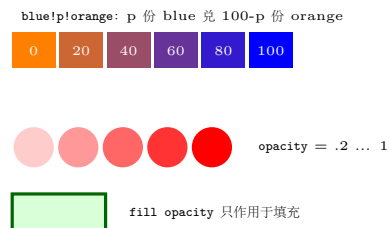
4 线宽与线型 line width · dash pattern · line cap

```
\begin{tikzpicture}[scale=.9,
  every node/.style={font=\tiny,text=gray,anchor=west}]
\foreach \s [count=\i] in {ultra thin,thin,thick,very thick,
  ultra thick,double}
  \draw[\s] (0,-.34*\i) -- (1.7,-.34*\i) node{\s};
\begin{scope}[xshift=4.1cm]
  \foreach \s [count=\i] in {dashed,dotted,densely dotted,
    dashdotted,loosely dashed}
    \draw[\s,thick] (0,-.34*\i) -- (1.7,-.34*\i) node{\s};
  \draw[thick,dash pattern=on 4pt off 1pt on 1pt off 1pt]
    (0,-2.04) -- (1.7,-2.04) node{dash pattern};
\end{scope}
\draw[line width=4pt,line cap=round,gray!50] (0,-2.6)--(.9,-2.6);
\draw[line width=4pt,line cap=butt,gray!80] (1.4,-2.6)--(2.3,-2.6);
\node at (2.4,-2.6) {line cap = round / butt};
\end{tikzpicture}
```



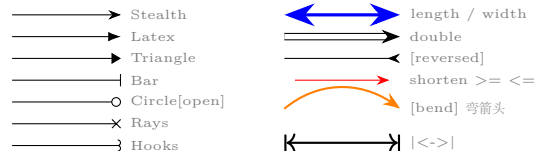
5 颜色与透明度 xcolor 混色 · opacity · fill opacity

```
\begin{tikzpicture}[scale=.95]
\foreach \p [count=\i from 0] in {0,20,...,100}
  \fill[blue!\p!orange] (\i*.66,0) rectangle ++(.6,.5)
  node[midway,font=\tiny,text=white]{\p};
\node[font=\tiny,anchor=west] at (0,.72)
  {\texttt{blue!\p!orange}: p 份 blue 兑 100-p 份 orange};
\foreach \o [count=\i from 0] in {.2,.4,.6,.8,1}
  \fill[red,opacity=\o] (\i*.62+.3,-1.1) circle (.28);
\node[font=\tiny,anchor=west] at (3.3,-1.1)
  {\texttt{opacity} = .2 ... 1};
\filldraw[fill=green!60,fill opacity=.25,line width=1.2pt,
  draw=green!40!black] (0,-2.3) rectangle (1.3,-1.75);
\node[font=\tiny,anchor=west] at (1.5,-2)
  {\texttt{fill opacity} 只作用于填充};
\end{tikzpicture}
```



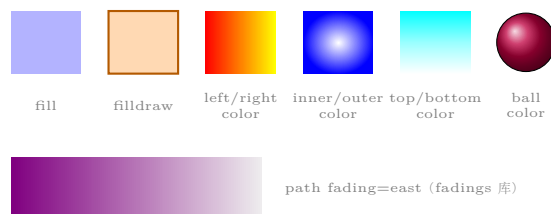
6 箭头 arrows.meta: 箭头族 · 参数化 · shorten

```
\begin{tikzpicture}[scale=.9,
  every node/.style={font=\tiny,text=gray,anchor=west}]
\foreach \a [count=\i] in {Stealth,Latex,Triangle,Bar,
  Circle[open],Rays,Hooks}
  \draw[-\a] (0,-.32*\i) -- (1.6,-.32*\i) node{\a};
\begin{scope}[xshift=4cm]
  \draw[blue,very thick,
    {Stealth[length=4mm,width=3mm]}--{Stealth[length=4mm]}]
    (0,-.32) -- (1.7,-.32) node{length / width};
  \draw[-{Stealth},double,double distance=2pt]
    (0,-.64) -- (1.7,-.64) node{double};
  \draw[-{Stealth[reversed]}] (0,-.96)--(1.7,-.96)
    node{[reversed]};
  \draw[red,-{Stealth},shorten >=4pt,shorten <=4pt]
    (0,-1.28) -- (1.7,-1.28) node{shorten >= <=};
  \draw[orange,thick,-{Stealth[bend]}]
    (0,-1.7) to[bend left=40] (1.7,-1.7) node{[bend] 弯箭头};
  \draw[|<->|,thick] (0,-2.2) -- (1.7,-2.2) node{|<->|};
\end{scope}
\end{tikzpicture}
```



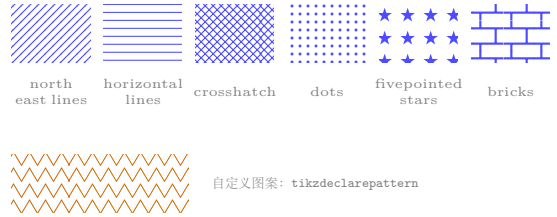
7 填充与渐变 even odd rule · shade · ball · fading

```
\begin{tikzpicture}[scale=.92,
  every node/.style={font=\tiny,text=gray}]
\fill[blue!30] (0,0) rectangle (1,.9);
\filldraw[fill=orange!30,draw=orange!70!black,thick]
  (1.4,0) rectangle (2.4,.9);
\shade[left color=red,right color=yellow] (2.8,0) rectangle (3.8,.9);
\shade[inner color=white,outer color=blue] (4.2,0) rectangle (5.2,.9);
\shade[top color=cyan,bottom color=white,middle color=cyan!40]
  (5.6,0) rectangle (6.6,.9);
\shadedraw[ball color=purple] (7.4,.45) circle (.42);
\fill[gray!12] (0,-2.1) rectangle (3.6,-1.2);
\fill[violet,path fading=east] (0,-2.1) rectangle (3.6,-1.2);
\foreach \x/\t in {5/fill, 1.9/filldraw, 3.3/{left/right}\color},
  4.7/{inner/outer}\color, 6.1/{top/bottom}\color,
  7.4/{ball}\color}
  \node[align=center,text width=1.3cm] at (\x,-.45) {\t};
\node[anchor=west] at (3.8,-1.65) {path fading=east (fadings 库)};
\end{tikzpicture}
```



8 图案填充 patterns · patterns.meta 自定义 tile

```
\begin{tikzpicture}[scale=.9]
\foreach \p [count=\i from 0] in {north east lines,
horizontal lines,crosshatch,dots,fivepointed stars,bricks}
{\fill[pattern=\p,pattern color=blue!70]
(\i*1.35,0) rectangle ++(1.15,.85);
\node[font=\tiny,text=gray,align=center,text width=1.4cm]
at (\i*1.35+.58,-.42) {\p};}
\tikzdeclarepattern{name=tri, tile size={(6pt,6pt)},
bottom left={(-.5pt,-.5pt)}, top right={(6.5pt,6.5pt)},
code={\draw[line width=.3pt] (0,0)--(3pt,5pt)--(6pt,0);}}
\fill[pattern=tri,pattern color=orange!80!black]
(0,-2.2) rectangle (2.6,-1.35);
\node[font=\tiny,text=gray,anchor=west] at (2.8,-1.78)
{自定义图案: \texttt{tikzdeclarepattern}};}
\end{tikzpicture}
```



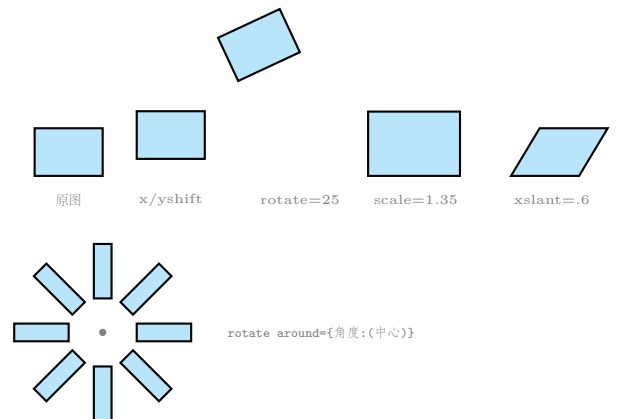
9 相对坐标 ++ 移动当前点 · + 不移动 · 命名坐标 · |- 组合

```
\begin{tikzpicture}[scale=.95]
\draw[thick,blue] (0,0) -- ++(1.2,0) -- ++(0,.9) -- ++(-1.2,0)
-- cycle;
\draw[thick,red] (2.4,0) -- +(1.2,0) (2.4,0) -- +(.6,.9)
(2.4,0) -- +(1.2,.9);
\coordinate (A) at (4.8,0); \coordinate (B) at (6.8,.9);
\draw[thick,orange] (A) -- (B);
\draw[gray,dashed] (A) -- (B |- A) -- (B); % (B|-A): B.x, A.y
\fill (A) circle(1.5pt) node[below]{$A$};
\fill (B) circle(1.5pt) node[above]{$B$};
\foreach \x/\t in {.6/{++(dx,dy) 链式}, 3/{+(dx,dy) 星形},
5.8/{(B |- A) 直角}}
\node[font=\tiny,text=gray] at (\x,-.42) {\t};
\end{tikzpicture}
```



10 坐标变换 shift · rotate · scale · xslant · rotate around

```
\begin{tikzpicture}[scale=.9,
blk/.style={draw,fill=cyan!25,thick},
lb/.style={font=\tiny,text=gray}]
\draw[blk] (0,0) rectangle (1,.7);
\draw[blk,xshift=1.5cm,yshift=.25cm] (0,0) rectangle (1,.7);
\draw[blk,rotate=25,xshift=3.3cm] (0,0) rectangle (1,.7);
\draw[blk,xshift=4.9cm,scale=1.35] (0,0) rectangle (1,.7);
\draw[blk,xshift=7cm,xslant=.6] (0,0) rectangle (1,.7);
\foreach \x/\t in {.5/{原图}, 2/{x/yshift}, 3.9/{rotate=25},
5.6/{scale=1.35}, 7.6/{xslant=.6}}
\node[lb] at (\x,-.35) {\t};
\begin{scope}[yshift=-2.3cm,xshift=1cm]
\foreach \a in {0,45,...,315}
\draw[blk,rotate around={\a:(0,0)}] (.5,-.13) rectangle (1.3,.13);
\fill[gray] (0,0) circle(1.5pt);
\node[lb,anchor=west] at (1.7,0)
{\texttt{rotate around={角度:(中心)}}};
\end{scope}
\end{tikzpicture}
```

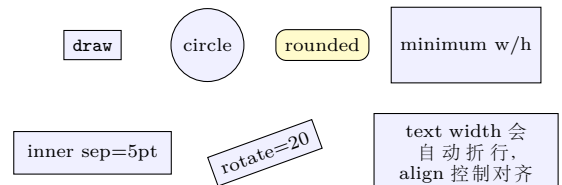


二 · 节点与标签

形状 / 锚点 / 定位 / 连线 / 矩阵 / 包围

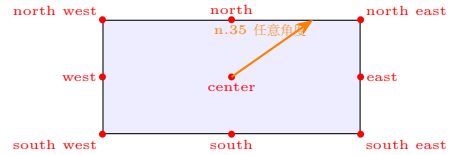
11 节点基础 draw · fill · minimum size · text width · align

```
\begin{tikzpicture}[scale=.95,
every node/.style={draw,fill=blue!6,font=\scriptsize}]
\node at (0,0) {\texttt{draw}};
\node[circle] at (1.6,0) {circle};
\node[rounded corners,fill=yellow!25] at (3.2,0) {rounded};
\node[minimum width=1.8cm,minimum height=1cm] at (5.2,0)
{minimum w/h};
\node[inner sep=5pt] at (0,-1.5) {inner sep=5pt};
\node[rotate=20] at (2.4,-1.5) {rotate=20};
\node[text width=2.2cm,align=center] at (5.2,-1.5)
{text width 会自动折行, align 控制对齐};
\end{tikzpicture}
```



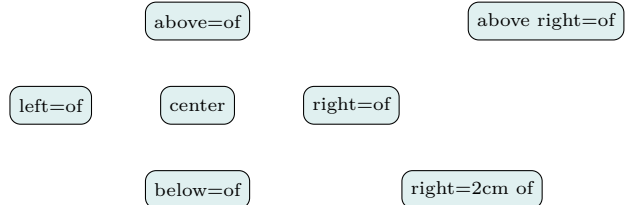
12 锚点 node.north / .south east / .center / . 任意角度

```
\begin{tikzpicture}
  \node[draw,minimum width=3.4cm,minimum height=1.5cm,
    fill=blue!7] (n) at (0,0) {};
  \foreach \a/\p in {north/above,south/below,east/right,west/left,
    north east/above right,north west/above left,
    south east/below right,south west/below left,center/below}
  {\fill[red] (n.\a) circle (1.3pt);
    \node[font=\tiny,text=red,\p=1pt,inner sep=1pt] at (n.\a) {\a};}
  \draw[-Stealth,orange,thick] (n.center) -- (n.35);
  \node[font=\tiny,text=orange,below left=-2pt and -1pt]
    at (n.35) {n.35 任意角度};
\end{tikzpicture}
```



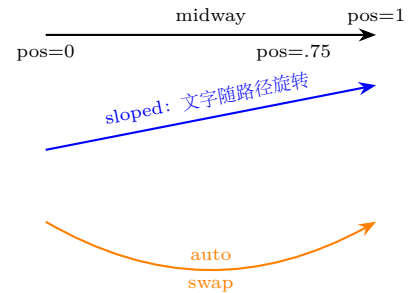
13 相对定位 positioning 库: above=of · node distance

```
\begin{tikzpicture}[node distance=6mm and 9mm,
  every node/.style={draw,rounded corners,fill=teal!12,
    font=\scriptsize,minimum size=5mm}]
  \node (c) {center};
  \node[above=of c] (a) {above=of};
  \node[below=of c] (b) {below=of};
  \node[left=of c] (l) {left=of};
  \node[right=of c] (r) {right=of};
  \node[above right=of r] (q) {above right=of};
  \node[right=2cm of b] (f) {right=2cm of};
\end{tikzpicture}
```



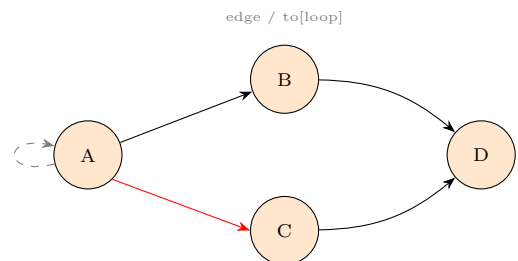
14 路径上的节点 pos · midway · sloped · auto · swap

```
\begin{tikzpicture}[scale=.95]
  \draw[-Stealth,thick] (0,0) -- (4.6,0)
    node[pos=0,below]{pos=0} node[midway,above]{midway}
    node[pos=.75,below]{pos=.75} node[above]{pos=1};
  \draw[-Stealth,thick,blue] (0,-1.6) -- (4.6,-.7)
    node[midway,sloped,above]{sloped: 文字随路径旋转};
  \draw[-Stealth,thick,orange] (0,-2.6) to[bend right=30]
    node[midway,auto]{auto} node[midway,auto,swap]{swap}
    (4.6,-2.6);
\end{tikzpicture}
```



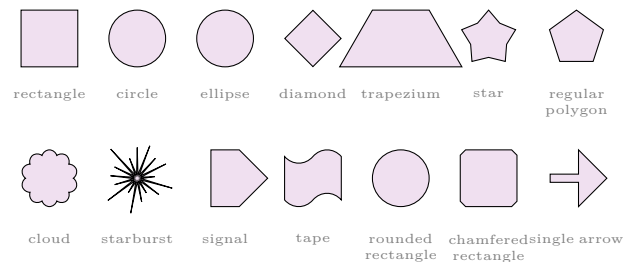
15 节点连线 边界自动裁剪 · edge · loop · 指定锚点

```
\begin{tikzpicture}[
  n/.style={draw,circle,fill=orange!20,minimum size=9mm}]
  \node[n] (A) at (0,0) {A};
  \node[n] (B) at (2.6,1) {B};
  \node[n] (C) at (2.6,-1) {C};
  \node[n] (D) at (5.2,0) {D};
  \draw[-Stealth] (A) -- (B); % 连线自动停在节点边界
  \draw[-Stealth,red] (A.south east) -- (C.west); % 指定锚点
  \path (B) edge[-Stealth,bend left=20] (D)
    (C) edge[-Stealth,bend right=20] (D);
  \draw[-Stealth,dashed,gray] (A) to[loop left] (A);
  \node[font=\tiny,text=gray,above] at (2.6,1.6) {edge / to[loop]};
\end{tikzpicture}
```



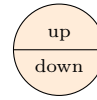
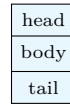
16 形状库 shapes.geometric / .symbols / .misc / .arrows

```
\begin{tikzpicture}[scale=.88,
  s/.style={draw,fill=violet!12,minimum size=7.5mm,inner sep=1pt},
  l/.style={font=\tiny,text=gray,align=center,anchor=north,
    text width=1.3cm}]
  \foreach \p [count=\i from 0] in {rectangle,circle,ellipse,diamond,
    trapezium,star,{regular polygon}}
  {\node[s,\p] at (\i*1.32,0) {}; \node[l] at (\i*1.32,-.62) {\p};}
  \foreach \p [count=\i from 0] in {cloud,starburst,signal,tape,
    {rounded rectangle},{chamfered rectangle},{single arrow}}
  {\node[s,\p] at (\i*1.32,-2.1) {}; \node[l] at (\i*1.32,-2.8) {\p};}
\end{tikzpicture}
```



17 多部分节点与标注 rectangle/circle split · label · pin

```
\begin{tikzpicture}
  \node[draw,rectangle split,rectangle split parts=3,
    fill=cyan!8,text centered] at (0,0)
    {head \nodepart{two} body \nodepart{three} tail};
  \node[draw,circle split,fill=orange!15] at (2.6,0)
    {up \nodepart{lower} down};
  \node[draw,fill=green!12,label=above:{label=above:...},
    pin=[pin edge={-Stealth,red}]-20:{pin: 带引线}]
    at (5.6,0) {node};
\end{tikzpicture}
```



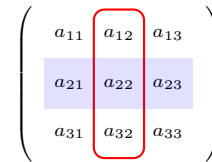
label=above:...

node

pin: 带引线

18 矩阵 matrix 库: matrix of math nodes · 定符号 · 单元引用

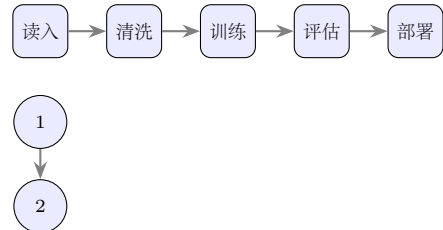
```
\begin{tikzpicture}
  \matrix (M) [matrix of math nodes, nodes={minimum size=6.5mm},
    left delimiter=(, right delimiter=)]
  {
    a_{11} & a_{12} & a_{13} \\
    a_{21} & a_{22} & a_{23} \\
    a_{31} & a_{32} & a_{33} \\
  };
  \draw[red,thick,rounded corners]
    (M-1-2.north west) rectangle (M-3-2.south east);
  \node[font=\tiny,text=red,below=1mm] at (M.south)
    {用 (M-行-列) 引用任意单元};
  \fill[blue,opacity=.12] (M-2-1.north west) rectangle (M-2-3.south east);
\end{tikzpicture}
```



用 (M-行-列) 引用任意单元

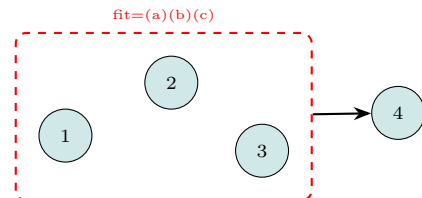
19 链式布局 chains 库: start chain · on chain · join

```
\begin{tikzpicture}[start chain=go going right, node distance=5mm,
  every on chain/.style={draw,rounded corners,fill=blue!8,
    minimum height=7mm,font=\scriptsize},
  every join/.style={-Stealth,thick,gray}]
\foreach \t in {读入,清洗,训练,评估,部署}
  \node[on chain=go,join] {\t};
\begin{scope}[start chain=dn going below,node distance=4mm]
  \node[on chain=dn,draw,circle,fill=orange!20] at (0,-1.2) {1};
  \node[on chain=dn,draw,circle,fill=orange!20,join] {2};
\end{scope}
\end{tikzpicture}
```



20 包围与分组 fit 库: 把若干节点框起来

```
\begin{tikzpicture}[
  n/.style={draw,circle,fill=teal!18,minimum size=7mm}]
\node[n] (a) at (0,0) {1};
\node[n] (b) at (1.4,.7) {2};
\node[n] (c) at (2.6,-.2) {3};
\node[n] (d) at (4.4,.3) {4};
\node[draw=red,thick,dashed,rounded corners,inner sep=3mm,
  fit=(a)(b)(c),label={[text=red,font=\tiny]above:%
  fit=(a)(b)(c)}] (grp) {};
\draw[-Stealth,thick] (grp) -- (d);
\end{tikzpicture}
```

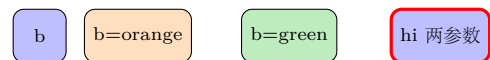


三 · 结构化与编程

样式 / 变换 / 循环 / 数学 / 坐标运算 / 图层

21 样式 / .style / .default / .style 2 args · every ...

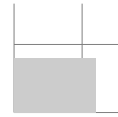
```
\begin{tikzpicture}[
  b/.style={draw,rounded corners,fill=#1!25,minimum size=7mm},
  b/.default=blue, % 不给参数时用它
  hi/.style 2 args={draw=#1,line width=#2},
  every node/.append style={font=\scriptsize}]
\node[b] at (0,0) {b};
\node[b=orange] at (1.3,0) {b=orange};
\node[b=green!70!black] at (3.3,0) {b=green};
\node[b,hi={red}{1.2pt}] at (5.3,0) {hi 两参数};
\tikzset{dot/.style={circle,fill=violet,inner sep=1.6pt}}
\foreach \i in {0,...,4} \node[dot] at (\i*.35,-1) {};
\node[font=\tiny,text=gray,anchor=west] at (1.6,-1)
  {\texttt{tikzset} 可在图中途定义样式};
\end{tikzpicture}
```



● ● ● ● ● tikzset 可在图中途定义样式

22 作用域 scope: 选项与变换只在组内生效, 且逐层累乘

```
\begin{tikzpicture}[scale=.9]
\draw[help lines] (0,0) grid (1.6,1.6);
\fill[gray!40] (0,0) rectangle (1.2,.8);
\begin{scope}[xshift=2.4cm,rotate=15,fill=orange!40]
\fill (0,0) rectangle (1.2,.8);
\begin{scope}[xshift=1.5cm,scale=.7,fill=blue!40]
\fill (0,0) rectangle (1.2,.8); % 变换在外层基础上叠加
\end{scope}
\end{scope}
\draw[red,thick,shift={(6.2,.4)},rotate=-25]
(0,0) -- (1.2,0) -- (1.2,.8) -- cycle;
\foreach \x/\t in {0.6/原位, 3.2/{scope 平移旋转},
6.6/{shift={(x,y)}}}
\node[font=\tiny,text=gray] at (\x,-.45) {\t};
\end{tikzpicture}
```



原位



scope 平移旋转



shift=(x,y)

23 循环 \foreach: 等差 · 多变量 · count · evaluate · 嵌套

```
\begin{tikzpicture}[scale=.9,
every node/.style={font=\tiny,text=gray,anchor=west}]
\foreach \x in {0,1,...,6} \fill[blue!50] (\x*.35,0) circle(.1);
\node at (2.6,0) {\{0,1,...,6\} 等差列表};
\foreach \x/\c in {0/red,1/orange,2/green!60!black,3/cyan}
\fill[\c] (\x*.35,-.5) circle(.1);
\node at (2.6,-.5) {\textbackslash x/\textbackslash c 多变量并行};
\foreach \t [count=\i] in {a,b,c,d}
\node[text=black,anchor=center] at (\i*.35-.35,-1) {\i\t};
\node at (2.6,-1) {[count=\textbackslash i] 计数};
\foreach \x [evaluate=\x as \y using {.4*sin(\x*36)}] in {0,...,10}
\fill[violet] (\x*.22,-1.9+\y) circle(.07);
\node at (2.6,-1.9) {[evaluate=\textbackslash x as \textbackslash y using
\> ...]];
\foreach \i in {0,...,3} \foreach \j in {0,...,2}
\fill[gray!70] (\i*.25,-2.8-\j*.25) circle(.07);
\node at (2.6,-3.05) {嵌套两层 = 网格};
\end{tikzpicture}
```

{0,1,...,6} 等差列表

\x/\c 多变量并行

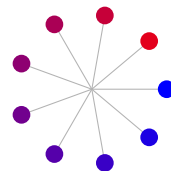
1a 2b 3c 4d [count=\i] 计数

[evaluate=\x as \y using ...]

嵌套两层 = 网格

24 数值计算 \pgfmathsetmacro · 函数库 · 随机数

```
\begin{tikzpicture}[scale=.9]
\pgfmathsetmacro{\n}{9} % 定义宏 (浮点)
\pgfmathsetmacro{\r}{1.1}
\foreach \i in {1,...,9}{
\pgfmathsetmacro{\a}{360/\n*\i}
\pgfmathsetmacro{\p}{100*\i/\n}
\draw[gray!60] (0,0) -- (\a:\r);
\fill[blue!\p!red] (\a:\r) circle (.13);}
\pgfmathsetmacro{\v}{veclen(3,4)} % = 5
\node[font=\tiny,text=gray,anchor=west,align=left] at (1.5,.3)
{\texttt{veclen(3,4)}=\v[.4mm]
常用函数: sin cos tan atan2 · sqrt pow exp ln\
· mod div int round · min max abs · pi e · rnd rand};
\foreach \i in {0,...,14}{
\pgfmathsetmacro{\h}{.15+.5*rnd} % 随机高度
\fill[teal!70] (1.6+\i*.16,-1.4) rectangle ++(.11,\h);}
\node[font=\tiny,text=gray,anchor=west] at (4.2,-1.2) {rnd 随机数};
\end{tikzpicture}
```

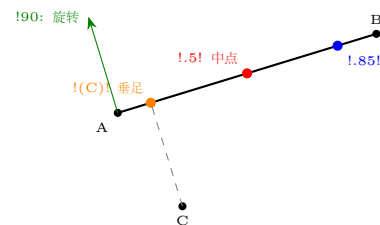


veclen(3,4)=4.99994
常用函数: sin cos tan atan2 · sqrt pow exp ln
· mod div int round · min max abs · pi e · rnd rand

rnd 随机数

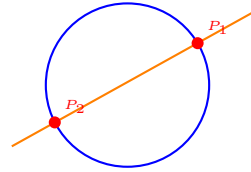
25 坐标运算 calc 库: \$(A)!(t)!(B)\$ 插值 · 投影 · 旋转

```
\begin{tikzpicture}[scale=.95, every node/.style={font=\tiny}]
\coordinate (A) at (0,0); \coordinate (B) at (3.6,1.1);
\coordinate (C) at (.9,-1.3);
\draw[thick] (A) -- (B);
\fill (A) circle(1.6pt) node[below left]{A};
\fill (B) circle(1.6pt) node[above]{B};
\fill (C) circle(1.6pt) node[below]{C};
\fill[red] ($(A)!.5!(B)$) circle(2pt) node[above left]{!.5! 中点};
\fill[blue] ($(A)!.85!(B)$) circle(2pt) node[below right]{!.85!};
\draw[dashed,gray] (C) -- ($(A)!(C)!(B)$);
\fill[orange] ($(A)!(C)!(B)$) circle(2pt)
node[above left=-1pt]{!(C)! 垂足};
\draw[green!50!black,-Stealth] (A) -- ($(A)!.4cm!90:(B)$)
node[left]{!90: 旋转};
\end{tikzpicture}
```



26 求交点 intersections 库: name path + name intersections

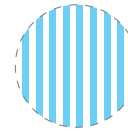
```
\begin{tikzpicture}[scale=.9]
\draw[name path=circ,thick,blue] (0,0) circle (1.2);
\draw[name path=line,thick,orange] (-1.7,-.9) -- (1.9,1.1);
\path[name intersections={of=circ and line,total=\n}]
\foreach \s in {1,...,\n}
{((intersection-\s) node[circle,fill=red,inner sep=1.6pt]}{
node[font=\tiny,text=red,above right=0pt]{$P_{\s}$}};
\node[font=\tiny,text=gray,anchor=west,align=left] at (2.2,0)
{先给路径起名 \texttt{name path=...}, \\\
再求交, 交点记作 \texttt{(intersection-i)}, \\\
\texttt{total=\textbackslash n} 回传交点个数。};
\end{tikzpicture}
```



先给路径起名 `name path=...`,
再求交, 交点记作 `(intersection-i)`,
`total=\n` 回传交点个数。

27 裁剪与叠加动作 clip · preaction · postaction

```
\begin{tikzpicture}[scale=.9]
\begin{scope}
\clip (0,0) circle (.9); % 之后的绘制被裁到圆内
\foreach \i in {0,...,9}
\fill[cyan!50] (-1+\i*.2,-1) rectangle ++(.1,2);
\end{scope}
\draw[gray,dashed] (0,0) circle (.9);
\draw[line width=3pt,blue,
postaction={draw=white,line width=1pt,dashed}]
(2.2,-.9) -- (2.2,.9); % 先粗后细 = 双色虚线
\fill[orange!70,
preaction={fill=black,opacity=.25,
transform canvas={xshift=2pt,yshift=-2pt}}]
(3.2,-.5) rectangle (4.4,.5); % 手工阴影
\foreach \x/\t in {0/clip, 2.2/postaction, 3.8/preaction}
\node[font=\tiny,text=gray] at (\x,-1.25) {\t};
\end{tikzpicture}
```



clip



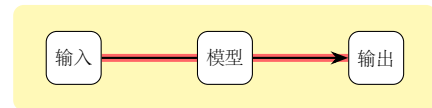
postaction



preaction

28 图层 backgrounds 库: on background layer 画到最底层

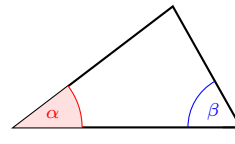
```
\begin{tikzpicture}[
n/.style={draw,fill=white,rounded corners,minimum size=7mm}]
\node[n] (a) at (0,0) {输入};
\node[n] (b) at (2,0) {模型};
\node[n] (c) at (4,0) {输出};
\draw[-Stealth,thick] (a)--(b) (b)--(c);
\begin{scope}[on background layer]
\fill[yellow!35,rounded corners]
(-.8,-.7) rectangle (4.8,.7);
\draw[red!60,line width=3pt] (a) -- (c);
\end{scope}
\node[font=\tiny,text=gray,below] at (2,-.9)
{黄底与粗红线虽后写, 却位于节点之下};
\end{tikzpicture}
```



黄底与粗红线虽后写, 却位于节点之下

29 可复用图元与角度标注 \pic · angles + quotes 库

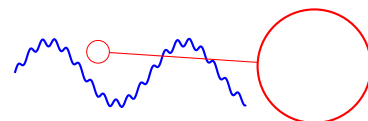
```
\begin{tikzpicture}[scale=1,
ra/.pic={\draw[line width=.6pt] (0,2.4mm) |- (2.4mm,0);}]
\coordinate (A) at (0,0); \coordinate (B) at (3,0);
\coordinate (C) at (2.1,1.6);
\draw[thick] (A) -- (B) -- (C) -- cycle;
\pic[draw=red,fill=red!12,angle radius=9mm,
"$\alpha$",text=red] {angle=B--A--C};
\pic[draw=blue,angle radius=7mm,"$\beta$",text=blue]
{angle=C--B--A};
\pic at (B) [rotate=180] {ra}; % 自定义 pic 复用
\node[font=\tiny,text=gray,anchor=west,align=left] at (3.4,.8)
{\texttt{\pic{angle=B--A--C}}} 自动画角\\
{\texttt{"$\alpha$"}} 由 quotes 库直接当标签\\
{\texttt{name/.pic={...}}} 定义可复用图元};
\end{tikzpicture}
```



`pic{angle=B--A--C}` 自动画角
"`$\alpha$`" 由 quotes 库直接当标签
`name/.pic={...}` 定义可复用图元

30 放大镜 spy 库: 局部放大并自动连线

```
\begin{tikzpicture}[scale=.9,
spy using outlines={circle,magnification=5,size=1.5cm,
connect spies,red}]
\draw[thick,blue,domain=0:3.4,samples=400,smooth]
plot (\x,{.45*sin(180*\x)+.06*sin(2000*\x)});
\spy on (1.1,.28) in node at (4.4,.1);
\end{tikzpicture}
```



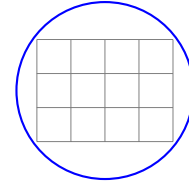
31 跨图引用 remember picture + overlay (需编译两次)

```
\begin{tikzpicture}[remember picture]
  \node[draw,fill=blue!12,rounded corners] (n1) {图 A 的节点};
\end{tikzpicture}\hspace{2.2cm}
\begin{tikzpicture}[remember picture]
  \node[draw,fill=orange!20,rounded corners] (n2) {图 B 的节点};
  \draw[overlay,-Stealth,red,thick] (n2) to[bend right=35] (n1);
\end{tikzpicture}
```



32 包围盒控制 use as bounding box · overlay 不占位

```
\begin{tikzpicture}[scale=.9]
  \draw[help lines,step=.5] (0,0) grid (2,1.5);
  \path[use as bounding box] (0,0) rectangle (2,1.5);
  \draw[thick,blue] (1,.75) circle (1.3); % 溢出但不撑大尺寸
  \node[font=\tiny,text=gray,anchor=west] at (2.6,.75)
    {蓝圆超出灰网格，但图的实际占位仍是网格大小};
\end{tikzpicture}
```



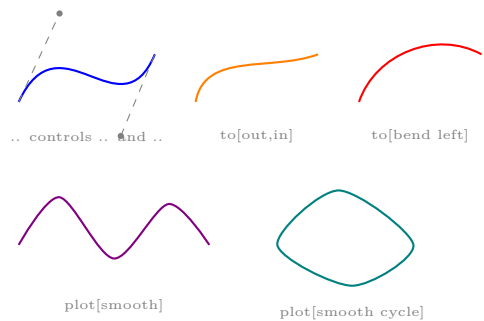
蓝圆超出灰网格，但图的实际占位仍是网格大小

四 · 曲线与装饰

贝塞尔 / 函数 / decorations / 阴影 / 混合模式

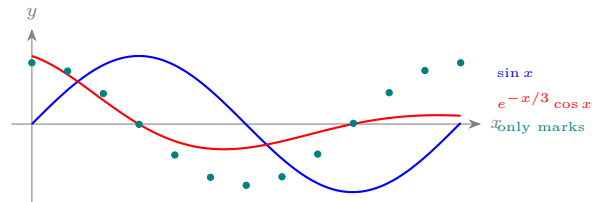
33 曲线族 controls · to[out,in] · bend · smooth · 闭合

```
\begin{tikzpicture}[scale=.9, every node/.style={font=\tiny, text=gray}]
  \draw[thick,blue] (0,0) .. controls (.6,1.3) and (1.5,-.5) .. (2,.7);
  \draw[gray,dashed] (0,0)--(.6,1.3) (1.5,-.5)--(2,.7);
  \fill[gray] (.6,1.3) circle(1.3pt) (1.5,-.5) circle(1.3pt);
  \draw[thick,orange] (2.6,0) to[out=80,in=200] (4.4,.7);
  \draw[thick,red] (5,0) to[bend left=50] (6.8,.7);
  \draw[thick,violet] plot[smooth] coordinates
    {(0,-2.1) (.6,-1.4) (1.4,-2.3) (2.2,-1.5) (2.8,-2.1)};
  \draw[thick,teal] plot[smooth cycle] coordinates
    {(3.8,-2.1) (4.7,-1.3) (5.8,-2.1) (4.9,-2.7)};
  \foreach \x/\y/\t in {1/-.5/{.. controls .. and ..},
    3.5/-.5/{to[out,in]}, 5.9/-.5/{to[bend left]},
    1.4/-3/{plot[smooth]}, 4.9/-3.1/{plot[smooth cycle]}}
    \node at (\x,\y) {\t};
\end{tikzpicture}
```



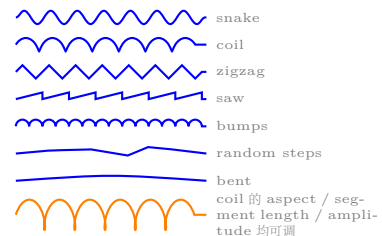
34 函数曲线 plot · domain · samples · 角度单位 r · mark

```
\begin{tikzpicture}[scale=.9]
  \draw[-Stealth,gray] (-.3,0) -- (6.6,0) node[right]{$x$};
  \draw[-Stealth,gray] (0,-1.2) -- (0,1.4) node[above]{$y$};
  \draw[thick,blue,domain=0:6.3,samples=120,smooth]
    plot (\x,{sin(\x r)}); % r: 把 \x 当弧度
  \draw[thick,red,domain=0:6.3,samples=120,smooth]
    plot (\x,{exp(-\x/3)*cos(\x r)});
  \draw[teal,domain=0:6.3,samples=13,only marks,mark=*,
    mark size=1.3pt] plot (\x,{.9*cos(\x r)});
  \node[font=\tiny,text=blue,anchor=west] at (6.7,.75) {$\sin x$};
  \node[font=\tiny,text=red,anchor=west] at (6.7,.35)
    {$e^{-x/3}\cos x$};
  \node[font=\tiny,text=teal,anchor=west] at (6.7,-.05)
    {only marks};
\end{tikzpicture}
```



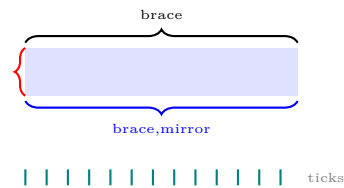
35 路径变形装饰 decorations.pathmorphing: decorate + decoration

```
\begin{tikzpicture}[scale=.9,
  every node/.style={font=\tiny, text=gray, right}]
  \foreach \d [count=i] in {snake,coil,zigzag,saw,bumps,
    random steps,bent}
    \draw[thick,blue,decorate,decoration={\d}]
      (0,-.4*i) -- (2.8,-.4*i) node{\d};
  \draw[thick,orange,decorate,
    decoration={coil,aspect=.4,segment length=4mm,amplitude=2mm}]
    (0,-3.3) -- (2.8,-3.3)
    node[align=left,text width=2.3cm]{coil 的 aspect /
      segment length / amplitude 均可调};
\end{tikzpicture}
```



36 路径替换装饰 decorations.pathreplacing: 花括号 brace · ticks

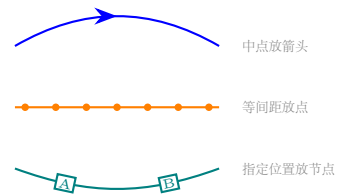
```
\begin{tikzpicture}[scale=.9]
\fill[blue!12] (0,0) rectangle (4,.7);
\draw[thick,decorate,decoration={brace,amplitude=5pt,raise=2pt}]
(0,.7) -- (4,.7) node[midway,above=7pt,font=\tiny]{brace};
\draw[thick,blue,decorate,
decoration={brace,mirror,amplitude=5pt,raise=2pt}]
(0,0) -- (4,0) node[midway,below=7pt,font=\tiny]
{brace,mirror};
\draw[thick,red,decorate,decoration={brace,amplitude=4pt},
rotate=90] (0,0) -- (.7,0);
\draw[decorate,decoration={ticks,segment length=8pt,amplitude=3pt},
thick,teal] (0,-1.2) -- (4,-1.2)
node[right,font=\tiny,text=gray]{ticks};
\node[font=\tiny,text=gray,anchor=west,align=left] at (4.6,-.4)
{\texttt{amplitude} 控制括号深度\\
\texttt{mirror} 翻到另一侧\\
\texttt{raise} 整体外移};
\end{tikzpicture}
```



amplitude 控制括号深度
mirror 翻到另一侧
raise 整体外移

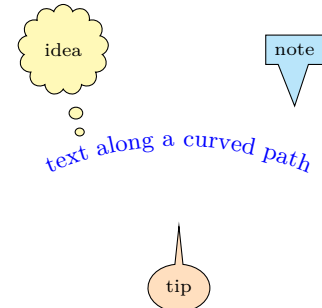
37 路径标记 decorations.markings: 中途箭头 · 等距分布

```
\begin{tikzpicture}[scale=.9]
\draw[thick,blue,postaction={decorate,decoration={markings,
mark=at position .5 with {\arrow[Stealth]{length=3mm}}}}]
(0,0) to[bend left=30] (3,0);
\draw[thick,orange,postaction={decorate,decoration={markings,
mark=between positions .05 and .95 step .15 with
{\fill circle(1.4pt);}}}]
(0,-.9) -- (3,-.9);
\draw[thick,teal,postaction={decorate,decoration={markings,
mark=at position .25 with {\node[transform shape,
fill=white,draw,inner sep=1pt,font=\tiny]{A};},
mark=at position .75 with {\node[transform shape,
fill=white,draw,inner sep=1pt,font=\tiny]{B};}}}]
(0,-1.8) to[bend right=20] (3,-1.8);
\foreach \y/\t in {0/{中点放箭头}, -.9/{等间距放点},
-1.8/{指定位置放节点}}
\node[font=\tiny,text=gray,right] at (3.2,\y) {\t};
\end{tikzpicture}
```



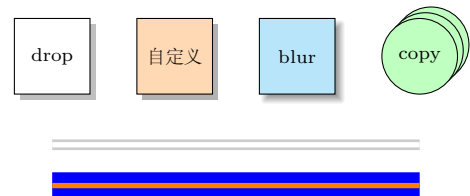
38 文字沿路径与气泡 decorations.text · shapes.callouts

```
\begin{tikzpicture}[scale=.9]
\path[decorate,decoration={text along path,
text={\small\color{blue}|text along a curved path|},
text align=center}] (0,-.3) to[bend left=30] (5.4,-.3);
\node[draw,cloud callout,cloud puffs=13,fill=yellow!35,
callout relative pointer={(.1,-.5)}] at (1,1.9) {idea};
\node[draw,rectangle callout,callout pointer width=4mm,
fill=cyan!25,
callout relative pointer={(0,-.55)}] at (4.4,1.9) {note};
\node[draw,ellipse callout,fill=orange!25,
callout relative pointer={(0,.5)}] at (2.7,-1.6) {tip};
\end{tikzpicture}
```



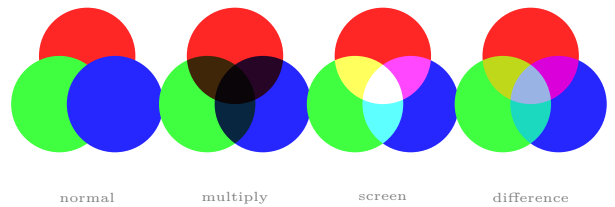
39 阴影与厚描边 shadows / shadows.blur · double · 描边叠加

```
\begin{tikzpicture}[scale=.9,
every node/.style={draw,minimum size=1cm,fill=white,
font=\scriptsize}]
\node[drop shadow] at (0,0) {drop};
\node[fill=orange!30,
drop shadow={shadow xshift=2pt,shadow yshift=-2pt,
opacity=.6}] at (1.8,0) {自定义};
\node[fill=cyan!25,blur shadow={shadow blur steps=10}]
at (3.6,0) {blur};
\node[circle,fill=green!25,double copy shadow] at (5.4,0) {copy};
\draw[line width=4pt,gray!40] (0,-1.3) -- (5.4,-1.3);
\draw[line width=2pt,white] (0,-1.3) -- (5.4,-1.3);
\draw[line width=4pt,double=orange,double distance=2pt,blue]
(0,-1.9) -- (5.4,-1.9);
\end{tikzpicture}
```



40 混合模式 blend group: multiply / screen / difference

```
\begin{tikzpicture}[scale=.85]
\foreach \m [count=\i from 0] in {normal,multiply,screen,difference}
{
\begin{scope}[xshift=\i*2.3cm,blend group=\m]
\fill[red!85] (90:.5) circle (.75);
\fill[green!75] (210:.5) circle (.75);
\fill[blue!85] (330:.5) circle (.75);
\end{scope}
\node[font=\tiny,text=gray] at (\i*2.3,-1.7) {\m};
}
\end{tikzpicture}
```

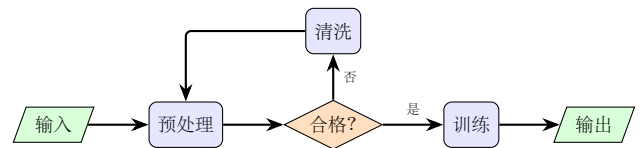


五 · 经典视觉呈现

流程图 / 网络 / 状态机 / 树 / 统计图 / 三维

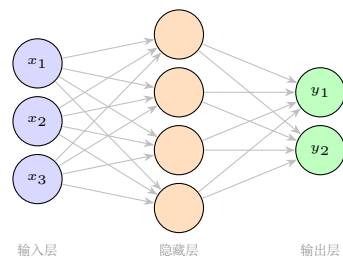
41 流程图 shapes.geometric + positioning + 折线连接

```
\begin{tikzpicture}[scale=.86,node distance=6mm and 8mm,
st/.style={draw,rounded corners=3pt,fill=blue!10,minimum height=6mm,
font=\scriptsize},
io/.style={draw,trapezium,trapezium left angle=70,
trapezium right angle=110,fill=green!15,font=\scriptsize},
dec/.style={draw,diamond,aspect=2,inner sep=1pt,fill=orange!25,
font=\scriptsize},
ar/.style={-Stealth,thick,rounded corners=2pt}]
\node[io] (a) {输入};
\node[st,right=of a] (b) {预处理};
\node[dec,right=of b] (c) {合格?};
\node[st,right=of c] (d) {训练};
\node[io,right=of d] (e) {输出};
\node[st,above=of c] (f) {清洗};
\draw[ar] (a)--(b); \draw[ar] (b)--(c);
\draw[ar] (c)--node[above,font=\tiny]{是}(d); \draw[ar] (d)--(e);
\draw[ar] (c)--node[right,font=\tiny]{否}(f);
\draw[ar] (f)--(b);
\end{tikzpicture}
```



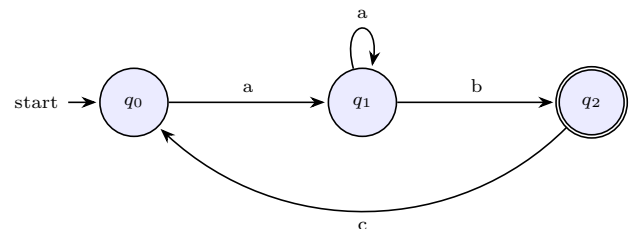
42 神经网络 foreach 双层循环生成全连接结构

```
\begin{tikzpicture}[scale=.85,
nn/.style={circle,draw,minimum size=6.5mm,inner sep=0pt,font=\tiny}]
\foreach \i in {1,2,3}
\node[nn,fill=blue!15] (i\i) at (0,-\i*.9+.9) {$x_{\i}$};
\foreach \j in {1,...,4}
\node[nn,fill=orange!25] (h\j) at (2.2,-\j*.9+1.35) {};
\foreach \k in {1,2}
\node[nn,fill=green!25] (o\k) at (4.4,-\k*.9+.45) {$y_{\k}$};
\foreach \i in {1,2,3} \foreach \j in {1,...,4}
\draw[gray!50,-{Stealth[length=1.4mm]}] (i\i) -- (h\j);
\foreach \j in {1,...,4} \foreach \k in {1,2}
\draw[gray!50,-{Stealth[length=1.4mm]}] (h\j) -- (o\k);
\foreach \x/\t in {0/输入层, 2.2/隐藏层, 4.4/输出层}
\node[font=\tiny,text=gray] at (\x,-2.9) {\t};
\end{tikzpicture}
```



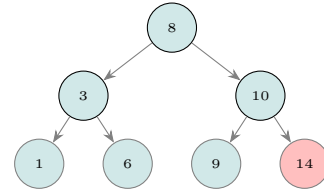
43 有限状态机 automata 库: state · initial · accepting · loop

```
\begin{tikzpicture}[scale=.9,>={Stealth[]},shorten >=1pt,auto,
node distance=2.1cm,semithick,
every state/.style={draw,fill=blue!8,minimum size=9mm,
font=\scriptsize}]
\node[state,initial] (q0) {$q_0$};
\node[state,right=of q0] (q1) {$q_1$};
\node[state,accepting,right=of q1] (q2) {$q_2$};
\path[->] (q0) edge node{a} (q1);
(q1) edge node{b} (q2);
(q1) edge[loop above] node{a} (q1);
(q2) edge[bend left=45] node[below]{c} (q0);
\end{tikzpicture}
```



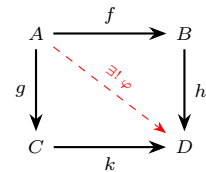
44 树 child 语法 · level distance · sibling distance

```
\begin{tikzpicture}[scale=.9,
  level distance=10mm,
  level 1/.style={sibling distance=26mm},
  level 2/.style={sibling distance=13mm},
  every node/.style={draw,circle,fill=teal!18,minimum size=6.5mm,
    inner sep=0pt,font=\tiny},
  edge from parent/.style={draw,gray,-Stealth}]
\node {8}
  child {node {3}}
  child {node {10}}
  child {node {6}}
  child {node {1}}
  child {node {9}}
  child {node[fill=red!25] {14}};
\end{tikzpicture}
```



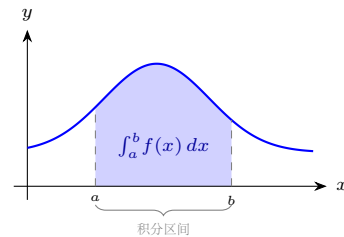
45 交换图 matrix + edge, 数学论文常用

```
\begin{tikzpicture}[scale=1]
\matrix (m) [matrix of math nodes, row sep=11mm, column sep=15mm]
{ A & B \\ C & D \\ };
\path[-Stealth,thick]
(m-1-1) edge node[above]{$f$} (m-1-2)
(m-1-1) edge node[left]{$g$} (m-2-1)
(m-1-2) edge node[right]{$h$} (m-2-2)
(m-2-1) edge node[below]{$k$} (m-2-2);
\draw[-Stealth,dashed,red] (m-1-1) --
node[sloped,above,font=\tiny]{$\exists!\varphi$} (m-2-2);
\end{tikzpicture}
```



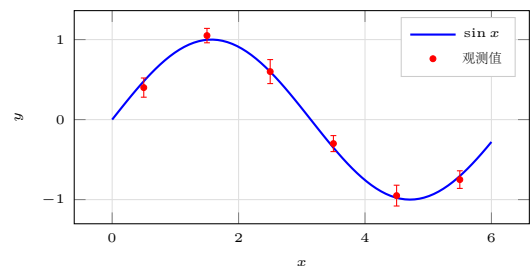
46 函数图与积分区域 plot + 闭合填充 + 花括号标注

```
\begin{tikzpicture}[scale=.9,
  declare function={f(\x)=.5+1.3*exp(-(\x-1.9)^2/1.2)};
\fill[blue!18,domain=1:3,samples=60,smooth]
  plot (\x,{f(\x)}) -- (3,0) -- (1,0) -- cycle;
\draw[-Stealth] (-.2,0) -- (4.4,0) node[right]{$x$};
\draw[-Stealth] (0,-.2) -- (0,2.3) node[above]{$y$};
\draw[thick,blue,domain=0:4.2,samples=80,smooth] plot (\x,{f(\x)});
\draw[dashed,gray] (1,0)--(1,{f(1)}) (3,0)--(3,{f(3)});
\node[font=\tiny,below] at (1,0) {$a$};
\node[font=\tiny,below] at (3,0) {$b$};
\node[font=\scriptsize,text=blue!60!black] at (2,.6)
{$\int_a^b f(x)\,dx$};
\draw[decorate,decoration={brace,mirror,amplitude=4pt,raise=7pt},
  gray] (1,0) -- (3,0)
node[midway,below=10pt,font=\tiny,text=gray]{积分区间};
\end{tikzpicture}
```



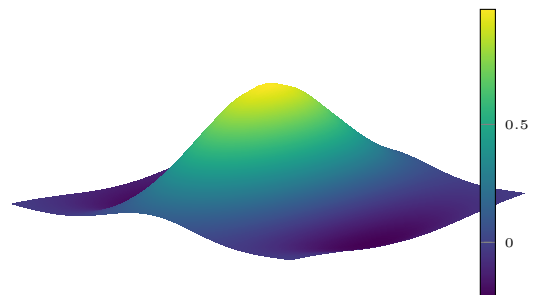
47 统计图 pgfplots: axis + addplot + 误差棒 + 图例

```
\begin{tikzpicture}
\begin{axis}[width=7.6cm,height=4.4cm,xlabel=$x$,ylabel=$y$,
  grid=major,grid style=gray!25,legend pos=north east,
  tick label style={font=\tiny},label style={font=\tiny},
  legend style={font=\tiny,draw=gray!40}]
\addplot[blue,thick,domain=0:6,samples=80]{sin(deg(x))};
\addplot[red,only marks,mark=*,mark size=1.1pt,
  error bars/.cd,y dir=both,y explicit]
  coordinates {(1.5,.4)+-(0,.12) (1.5,1.05)+-(0,.09)
    (2.5,.6)+-(0,.15) (3.5,-.3)+-(0,.1)
    (4.5,-.95)+-(0,.13) (5.5,-.75)+-(0,.11)};
\legend{$\sin x$,观测值}
\end{axis}
\end{tikzpicture}
```



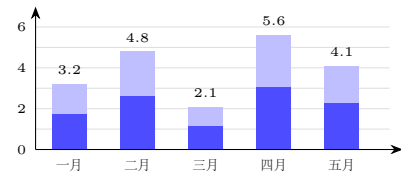
48 三维曲面 pgfplots: surf + colormap + 等高线投影

```
\begin{tikzpicture}
\begin{axis}[width=8.4cm,height=5.4cm,view={40}{28},
  colormap/viridis,ticks=none,axis lines=none,
  colorbar,colorbar style={width=2mm,xshift=-9mm,
    ticklabel style={font=\tiny}}]
\addplot3[surf,shader=interp,domain=-2:2,domain y=-2:2,
  samples=28,samples y=28]
  {exp(-(x^2+y^2)/2)*cos(deg(1.6*x))};
\end{axis}
\end{tikzpicture}
```



49 条形图 纯 TikZ: foreach 驱动数据 + 背景网格

```
\begin{tikzpicture}[scale=.9]
\foreach \y in {1,...,6} \draw[gray!25] (0,\y*.3) -- (5.2,\y*.3);
\foreach \v/\t [count=\i from 0] in
{3.2/一月,4.8/二月,2.1/三月,5.6/四月,4.1/五月}{
\fill[blue!25] (\i*.25,0) rectangle ++(.5,\v*.3);
\fill[blue!70] (\i*.25,0) rectangle ++(.5,{\v*.3*.55});
\node[font=\tiny,below] at (\i*.25,0) {\t};
\node[font=\tiny,above] at (\i*.25,\v*.3) {\v};}
\draw[-Stealth] (0,0) -- (5.4,0);
\draw[-Stealth] (0,0) -- (0,2.1);
\foreach \y in {0,2,4,6}
\node[font=\tiny,left] at (0,\y*.3) {\y};
\end{tikzpicture}
```



50 热力图 嵌套 foreach 读二维列表 + 颜色映射

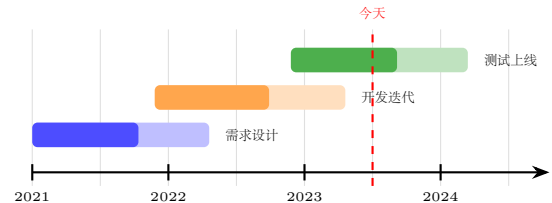
```
\begin{tikzpicture}[scale=.9]
\foreach \row [count=\i from 0] in
{{88,4,3,5},{6,79,9,6},{3,7,85,5},{4,5,6,85}}{
\foreach \v [count=\j from 0] in \row {
\fill[blue!\v!white] (\j*.62,-\i*.62) rectangle ++(.6,-.6);
\pgfmathtruncatemacro{\g}{\v>50 ? 0 : 100} % 深底配白字
\node[font=\tiny,text=black!\g]
at (\j*.62+.3,-\i*.62-.3) {\v};}}
\foreach \t [count=\j from 0] in {猫,狗,鸟,鱼}{
\node[font=\tiny,above] at (\j*.62+.3,.05) {\t};
\node[font=\tiny,left] at (-.05,-\j*.62-.3) {\t};}
\node[font=\tiny,text=gray,anchor=west] at (3,-1.2)
{混淆矩阵: \texttt{blue!\textbackslash v!white} 把数值直接当混色比例};
\end{tikzpicture}
```

	猫	狗	鸟	鱼
猫	88	4	3	5
狗	6	79	9	6
鸟	3	7	85	5
鱼	4	5	6	85

混淆矩阵: blue!\v!white 把数值直接当混色比例

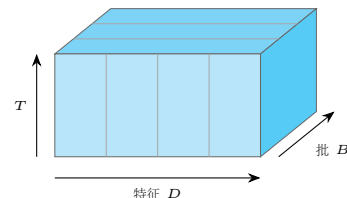
51 时间线与甘特 坐标即时间, foreach 批量画条

```
\begin{tikzpicture}[scale=.9]
\foreach \x in {0,1,...,7} \draw[gray!25] (\x,-.2) -- (\x,2.1);
\draw[thick,-Stealth] (0,0) -- (7.6,0);
\foreach \x/\t in {0/2021,2/2022,4/2023,6/2024}{
\draw[thick] (\x,.12) -- (\x,-.12);
\node[font=\tiny,below] at (\x,-.15) {\t};}
\foreach \s/\e/\y/\c/\t in {0/2.6/.55/blue/需求设计,
1.8/4.6/1.1/orange/开发迭代, 3.8/6.4/1.65/green!55!black/测试上线}{
\fill[\c!25,rounded corners=2pt] (\s,\y-.18) rectangle (\e,\y+.18);
\fill[\c!70,rounded corners=2pt] (\s,\y-.18)
rectangle ({\s+(\e-\s)*.6},\y+.18);
\node[font=\tiny,right] at (\e+.1,\y) {\t};}
\draw[red,thick,dashed] (5,-.2) -- (5,2.1)
node[above,font=\tiny,text=red]{今天};
\end{tikzpicture}
```



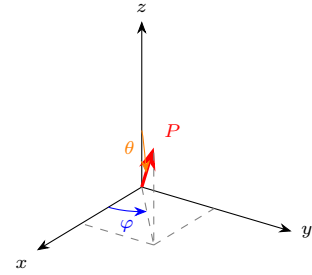
52 三维方块 自定义 x/y/z 单位向量做斜投影, 画张量

```
\begin{tikzpicture}[scale=.8, % z 轴指向右上
x={(1cm,0cm)}, y={(0cm,1cm)}, z={{.42cm,.34cm}}],
f/.style={draw=black!60,line width=.3pt}]
\def\X{3.4} \def\Y{1.7} \def\Z{2.2}
\fill[f,fill=cyan!45] (0,\Y,0)--(\X,\Y,0)--(\X,\Y,\Z)--(0,\Y,\Z)
--cycle; % 顶面
\fill[f,fill=cyan!55] (\X,0,0)--(\X,\Y,0)--(\X,\Y,\Z)--(\X,0,\Z)
--cycle; % 右侧面
\fill[f,fill=cyan!25] (0,0,0)--(\X,0,0)--(\X,\Y,0)--(0,\Y,0)
--cycle; % 正面
\foreach \i in {1,...,3}{ % 正面切分: 特征维
\draw[black!35,line width=.2pt] (\i*\X/4,0,0)--(\i*\X/4,\Y,0);}
\foreach \j in {1,2} % 顶面切分: 批
\draw[black!35,line width=.2pt] (0,\Y,\j*\Z/3)--(\X,\Y,\j*\Z/3);
\draw[-Stealth] (0,-.35,0)--(\X,-.35,0)
node[midway,below,font=\tiny]{特征 $D$};
\draw[-Stealth] (-.3,0,0)--(-.3,\Y,0)
node[midway,left,font=\tiny]{特征 $T$};
\draw[-Stealth] (\X+.3,0,0)--(\X+.3,0,\Z)
node[midway,below right,font=\tiny]{批 $B$};
\end{tikzpicture}
```



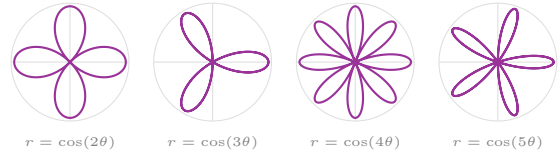
53 三维坐标系 tikz-3dplot: 球坐标 · 角度弧线

```
\tdplotsetmaincoords{65}{125}
\begin{tikzpicture}[tdplot_main_coords,scale=1.1]
\draw[-Stealth] (0,0,0)--(2,2,0,0) node[below left] {$x$};
\draw[-Stealth] (0,0,0)--(0,2,2,0) node[right] {$y$};
\draw[-Stealth] (0,0,0)--(0,0,2,2) node[above] {$z$};
\tdplotsetcoord{P}{2.1}{52}{40} % r, theta, phi
\draw[-Stealth,very thick,red] (0,0,0)--(P) node[above right]{$P$};
\draw[dashed,gray] (0,0,0)--(Pxy)--(P);
\draw[dashed,gray] (Pxy)--(Px) (Pxy)--(Py);
\tdplotdrawarc[blue,-Stealth]{(0,0,0)}{.7}{0}{40}
{anchor=north}{$\varphi$}
\tdplotsetthetaplanecoords{40}
\tdplotdrawarc[tdplot_rotated_coords,orange,-Stealth]{(0,0,0)}{.75}
{0}{52}{anchor=east}{$\theta$}
\end{tikzpicture}
```



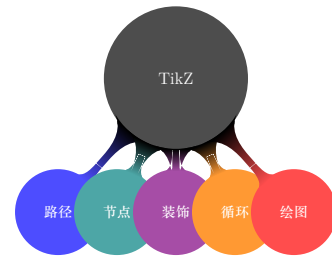
54 极坐标曲线 plot + variable + 极坐标写法 (角度: 半径)

```
\begin{tikzpicture}[scale=.82]
\foreach \k [count=\i from 0] in {2,3,4,5}{
\begin{scope}[shift=({\i*2.3,0})]
\draw[gray!25] (0,0) circle (.95);
\draw[gray!25] (-.95,0)--(.95,0) (0,-.95)--(0,.95);
\draw[thick,violet!80,domain=0:360,samples=400,smooth,
variable=\t] plot ({\t}:{.9*cos(\k*\t)});
\node[font=\tiny,text=gray] at (0,-1.3) {$r=\cos(\k\theta)$};
\end{scope}
}
\end{tikzpicture}
```



55 思维导图 mindmap 库: concept · 自动配色与连接

```
\begin{tikzpicture}[mindmap,scale=.52,transform shape,
every node/.style={concept,text=white,font=\bfseries},
root concept/.append style={concept color=black!70,
minimum size=2.2cm},
level 1 concept/.append style={level distance=3.4cm,
sibling angle=72,minimum size=1.8cm,font=\small}]
\node[root concept] {TikZ}
child[concept color=blue!70] {node[concept] {路径}}
child[concept color=teal!70] {node[concept] {节点}}
child[concept color=violet!70] {node[concept] {装饰}}
child[concept color=orange!80] {node[concept] {循环}}
child[concept color=red!70] {node[concept] {绘图}};
\end{tikzpicture}
```



附录 · 速查

库清单 / 常见陷阱 / 编译与导出

常用库清单 (\usetikzlibrary{a,b,...}, pgfplots 与 tikz-3dplot 是独立宏包, 用 \usepackage)

箭头 | arrows.meta (Stealth Latex Bar Circle Rays Hooks, 可带 [length= width= reversed open bend]), bending (箭头随曲线弯曲)
坐标 | calc (\$(A)!t!(B)\$ 插值/投影/旋转)、intersections (求交)、through (过点画圆)、positioning (above-of)
形状 | shapes.geometric.symbols.misc.multipart.arrows.callouts
结构 | scopes fit matrix chains backgrounds spy folding
装饰 | decorations.pathmorphing (snake/coil/zigzag)、.pathreplacing (brace)、.markings (路径中途放箭头/节点)、.text (文

字沿路径)、.shapes
填充 | patterns patterns.meta fadings shadows shadows.blur
数学与绘图 | math (\tikzmath)、plotmarks、datavisualization
三维 | 3d (canvas is xy plane at z=)、perspective (3d view={方位角}{仰角})、宏包 tikz-3dplot、pgfplots
专题 | automata (状态机)、petri (Petri 网)、trees mindmap graphs calendar、angles quotes (角度标注)、babel (与中文/语言包共存)、external (图形外部化缓存)

十个常见陷阱

- 1 每条路径必须以分号 ; 结尾, \node 也是路径, 漏分号是最高频错误。
- 2 scale=2 只放大坐标, 不放大线宽与文字; 要整体等比放大用 transform shape, 或改单位向量 x=2cm,y=2cm。
- 3 (a) -- (b): 若 a、b 是节点名, 连线自动止于边界; 若是坐标则直连中心, 两者不可混淆。
- 4 三角函数默认吃角度: sin(90)=1; 要按弧度须写 sin(\x r)。
- 5 选项里的逗号会切分键值, 含逗号的值要包一层 {}:
label={ [red]above:\$x,y\$}。
- 6 \foreach 的循环变量、\pgfmathsetmacro 的宏都是局部的, 出组即失效 (name intersections of the total= 宏同理)。
- 7 remember picture/overlay 依赖 .aux, 必须编译两次才正确。
- 8 节点文字大小跟随正文字号; 统一图内字号用 every picture/.append style={font=\small}。
- 9 pattern 与 shading 不能同时作用于同一次填充, 需分两条路径叠加。
- 10 中文必须用 XeLaTeX/LuaLaTeX + ctex (或 xeCJK), 并加载 babel 库以免与语言包冲突。

编译 · 外部化 · 导出

单张图独立成文件 (裁剪到内容边界, 便于插入他处):

```
\documentclass[tikz,border=2mm]{standalone}
\usetikzlibrary{arrows.meta}
\begin{document}
\begin{tikzpicture} \draw[-Stealth] (0,0)--(2,1); \end{tikzpicture}
\end{document}
```

编译: xelatex f.tex (中文) / pdflatex (纯英文更快) / lualatex (graphdrawing 自动布局必须用它); latexmk -pdfxe f.tex 自动跑够次数。

图多编译慢: 外部化把每张图缓存成独立 PDF, 只重编改动过的那张:

```
\usetikzlibrary{external}
\tikzexternalize[prefix=figs/] % 需 -shell-escape
```

导出位图/矢量:

```
pdftocairo -png -r 300 -singlefile fig.pdf fig # 位图
pdftocairo -svg fig.pdf fig.svg # 矢量
dvisvgm --pdf --font-format=woff fig.pdf # 字体转曲
```